

Full Project Proposal

Project name:Egremengead Crowdsourced Urban Logistics Platform

Business model:Shared Logistics Platform (Asset-Light Platform Model)

Main purpose:Reducing the cost of delivery within the city, converting existing travelers/youth trips into revenue without increasing traffic congestion.

Section 1: Executive Summary

1.1. Introduction and Vision of the Project

Street deliveryIt is an innovative project designed to transform the entire city's logistics and delivery network in Addis Ababa into an efficient and profitable shared economy by leveraging technology.

The main objective of this project is to enable the organization to operate without purchasing any fleet of vehicles and without allocating significant capital.**Asset-Light Platform Model**)—Accommodating thousands of pedestrians, travelers, unemployed youth, and small drivers who move from place to place in the city every day for their own business, work, education, or entertainment.**The first 4-Tier Grid Crowdsourced Logistics Platform in our country**It is to be.

1.2. Current Delivery Market Survey of Our Country (Local Market Reality)

The current delivery system in Addis Ababa and other major cities is not in line with the growth of the city and the needs of the people. The market is plagued by three major overlapping problems:

- **Exorbitant Delivery Cost:**The current traditional couriers or delivery companies charge between 250 and 500 birr for a small document or parcel sent over short distances (for example, from Kazanchis to Bole). This price is completely unaffordable for small traders, tenants, and the middle class, forcing people to travel by taxi themselves.

- **Traffic congestion and environmental toll:**In a city where tens of thousands of public transport users and pedestrians travel empty-handed every day, the addition of hundreds of additional motorbikes and cars to the roads solely for the purpose of transporting goods exacerbates the city's traffic congestion and smog (carbon emissions).
- **One-Size-Fits-All Approach:**A simple legal document worth 50 birr, a bill worth 50,000 birr, and household goods worth 3,000 birr are all required to be transported by motorcycle or similar means of transport. The lack of a tiered payment system that takes into account the value, type, security, and time requirements of the goods creates security risks and wastes money.

1.3. Strategies taken from international experiences (Global Benchmarks & Lessons)

Street Delivery combines three major logistics models that have been successful in developed and developing countries, while taking into account the social and economic realities of our country:

- **Crowdsourcing Model — Instacart & Roadie:**By taking the approach of ordinary people in America and Europe who can earn extra income by delivering goods along their regular routes without changing their daily commute, it will transform young people and students who commute by taxi, train, and on foot in Addis Ababa into a productive workforce.
- **Multi-tiered service structure (Tiered Dynamic Matching — Lalamove):**By adopting the successful tiered delivery algorithm that has been used in Asian cities, from small parcels to large trucks, customers can choose according to their budget and the nature of the item.
- **Strict Security and Delivery Chain (Digital Chain of Custody — Glovo & Uber Connect):**To address the risk of loss and breakage of items, the system has fully integrated the 2-step OTP secret number, Live GPS Tracking, and Photo Verification system used by international organizations.

1.4. The Strategic Solution for Street Delivery

This project aims to overcome the aforementioned problems.**4-Tier Smart Logistics System**It will come with:

[Step 1: Express] → On foot / by taxi (documents, clothes, small items)

[Level 2: Standard] → By bike / motorbike (food, medicine, cake)

[Step 3: Secure] → Wallet Hold (Expensive Electronics)

[Step 4: Cargo] → Suzuki / Isuzu (Home and Office Goods, Cargo)

Based on this structure, the app connects the customer with a pedestrian or driver traveling on that route within seconds, based on the level selected by the customer, the type of item, and the destination.

1.5. Summary of Social and Economic Impact (Impact & ROI)

Street delivery is not only a business but also a social enterprise project with great social benefits:

- **Job creation:**It will enable thousands of unemployed youth and students living in Addis Ababa to earn their own income through their daily commute, without any initial capital or vehicle.
- **Cost reduction:**Small traders and households can reduce their delivery costs**60% to 70%**They allow them to reduce.
- **Low Financial Risk and Fast Implementation:**Since the organization does not purchase cars, motorcycles, or warehouses**850,000 Ethiopian Birr**With minimal initial capital, it can build its app, call center, and robust security infrastructure and be fully operational within 5 months.

Part 2: Problem Statement & Market Gap

2.1. Introduction and the overall reality of the city

Although Addis Ababa, the capital of Ethiopia, is one of the fastest growing cities in Africa, the city's infrastructure development has not kept pace with the growth of its population and economic activity, putting a huge strain on the logistics sector. The current urban last-mile logistics system in the city is fragmented, expensive, time-consuming, and does not follow modern mapping.

2.2. Analysis of the main problems of the root cause

1. Exorbitant & Volatile Pricing

Currently, traditional motorbike deliveries and small delivery companies operating in the city do not have a fixed and clear price range.

- **Cost of small document/item:**A small trader or individual is forced to pay 250 to 400 birr to send a document weighing less than 100 grams from Piyasa to Bole (about 10 kilometers).
- **Economic pressure:**Since this fee is completely disproportionate to the cost of the item, it causes small online merchants (Social Media Vendors) to lose their profits or lose their customers.
- **The price of one's own self:**A system has developed where motorists take advantage of rainy conditions, traffic congestion, or lack of time to increase prices at will.

2. Urban Congestion & Resource Inefficiency

Every day in Addis Ababa city, thousands of workers, students, and travelers move from place to place by public transportation (taxi, train, Sheger/Anbessa bus) or on foot.

- **Empty Trips:**These thousands of people travel the same route every day, empty-handed, without any belongings.
- **Overlapping traffic pressure:**On the other hand, hundreds of additional motorists and vehicles engaged solely in delivering goods will use the same

route, causing traffic congestion, fuel waste, and air pollution. The lack of a crowdsourcing system for logistics is a huge waste of resources.

[Current situation] → People travel empty-handed + more cars on the road = more traffic and pollution

[Pedestrian] → Using existing pedestrian paths = zero additional traffic + low cost

3. Lack of Tiered Categorization

Current delivery services handle all items in the same way (One-Size-Fits-All).

- **Lack of caution and price discrimination:**A plain piece of paper worth 50 birr and a laptop worth 80,000 birr are packed into the trunk of a motorbike.
- **Specific option:**The customer does not have the right to choose whether his item is expensive and should be delivered with special care and insurance, or whether his item is ordinary and should be delivered by a cheap person with a convenient card.

4. Security & Trust Deficit

The current system does not have a digital chain of custody for ensuring the safety of goods.

- **Risk of breakage and loss:**Once the item is delivered to the addressee, there is no live tracking or one-time password (OTP) to confirm the status of the item, where it will arrive, and whether it was delivered to the right person.
- **Lack of accountability:**If an item is lost or broken, the warranty given to the addressee is limited to verbal or paper identification, so it takes a long time to legally claim and obtain compensation.

2.3. Market Gap Analysis

There are currently three major unfilled gaps in the logistics market in Addis Ababa:

+-----+

| Unfilled market gaps |

+-----+

| 1. The Micro-Vendors Gap |

| - Small merchants selling on TikTok/Telegram are looking for cheap shipping. |

+-----+

| 2. Commuter Monetization Gap |

| - Students and travelers want to deliver goods to cover their transportation costs. |

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| 3. Cargo & Relocation Gap |

| - Lack of a Suzuki/Isuzu cargo app with clear pricing and mapping. |

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- **Micro-Vendors Gap:** There are thousands of young traders selling clothes, shoes, and small electronics on Telegram, Instagram, and TikTok. These traders cannot afford to pay a delivery fee of 300 birr because their profit does not exceed 100–200 birr. They want a cheaper delivery method that costs 50–80 birr.
- **Uncaptured Commuter Labor Gap:** No digital platform has been developed to enable tens of thousands of college students, office workers, and unemployed youth in the city to turn their daily commute into income.
- **Cargo Logistics Gap:** Most current apps are focused on food and small-motor vehicle deliveries. Customers looking to move furniture, transport

office equipment, and transport large commercial cartons in Suzuki, Piaggio, or Isuzu are still forced to negotiate high prices with drivers in their neighborhoods.

2.4. Summary Comparison (Current Market vs. Egremengead Solution)

Measurement criteria	Current delivery procedure	"On-the-go delivery" solution
Service fee	High and approximate (250 - 500+ birr)	Very cheap and calculated by map (starting at 60 birr)
Vehicle use	Only separate engines (creates additional traffic)	Using existing travelers and pedestrians (does not increase traffic)
Tiers	No (same procedure for all items)	4-tier system (Express, Standard, Secure, Cargo)
safety assurance	Only by verbal and paper identification	2-Step OTP, Wallet Hold, and Live GPS Viewing
Traveler/Youth Engagement	Only some of the dead	Any traveler/student with a Fayda ID can participate.

Part 3: Analysis of Global Best Practices & Benchmarks

3.1. Introduction

Crowdsourced Logistics is a rapidly growing sector worldwide. "Crowd-sourced delivery" has gradually integrated three major international practices that have been highly successful in North America, Asia, and Europe, while taking into account the social and economic conditions of our country.

3.2. International experiences compiled

1. Instacart & Roadie (USA) — Crowdsourced Commuter Delivery

- **The procedure:** Ordinary people can earn extra income by delivering goods or purchases along their route, without changing their daily commute.
- **Lessons learned from them:** Rather than expecting people to make a specific journey, it has been found that route-matching items using their regular origin-destination route can significantly reduce costs and time.

2. Lalamove (Asia / Thailand / Philippines) — Multi-Tier Logistics Grid

- **The procedure:** From small documents to large trucks (Pickups/Trucks), an automatic pricing system based on vehicle type and weight, divided into tiers in one application.
- **Lessons learned from them:** Allowing the customer to choose the vehicle and price level that best suits the size and urgency of the item will increase market acceptance.

3. Glovo & Uber Connect (Europe & Africa) — Security & Chain of Custody

- **The procedure:** A 4-digit one-time password (OTP) and live GPS tracking system that provides a phone application to prevent items from getting lost or misplaced on the road.

- **Lessons learned from them:**Public trust will be built quickly if digital delivery verification and multi-level security are implemented.

Part 4: Tiered Delivery Architecture

The main strength of "On-Delivery" is that it organizes goods into 4 strict tiers, based on price, type, delivery speed, and transportation methods.

Level 1: "Pedestrian Express" (Micro & Commuter Delivery)

- **Type of item:**Documents, envelopes, keys, clothing, small pharmacy supplies.
- **The highest price of the item:**Up to 3,000 birr.
- **Delivery speed:**Within 2 to 4 hours (depending on the route of the traveler).
- **Transportation method:**Ordinary travelers, students, and unemployed youth traveling by foot, train, public taxi, or rideshare.
- **Security:**Photo verification and 4-digit OTP.

Level 2: "On-Road Standard" (Direct Bike/Motor Delivery)

- **Type of item:**Fresh foods, cakes, medicines, urgent business items.
- **The highest price of the item:**Up to 7,000 birr.
- **Delivery speed:**Fast/urgent (30 to 60 minutes).
- **Transportation method:**By bicycle, electric scooter, and motorcycle.
- **Security:**Thermal Bag + GPS tracking.

Level 3: "En-route Secure" (High-Value / Sensitive Delivery)

- **Type of item:**Smartphones, laptops, expensive jewelry, confidential legal/bank documents.
- **The highest price of the item:**From 8,000 to 100,000 birr.
- **Delivery speed:**Direct Courier.
- **Transportation method:**Highly trusted selected motorists or private small cars (Vitz/Demio).

- **Security: Wallet Hold System**— The recipient cannot receive the item unless the amount of money equivalent to the estimated value of the item is locked in the App Wallet.

Level 4: "Heavy & Relocation Cargo"

- **Type of item:**Household and office supplies, cartons that businesses ship in bulk, construction samples.
- **Price of the item:**Unlimited (covered by insurance).
- **Delivery speed:**At the appointed time (Scheduled).
- **Transportation method:**Suzuki Drive (Carry), Piaggio, Isuzu (Dyna/FSR).
- **Security:**Trained installers + cardboard/bubble wrap packaging.

Pricing Algorithm Formula

The total cost of a delivery is calculated by the system using the following formula:

$$\{\text{Total Price}\} = \{\text{Base Fare}\} + \{\text{Distance in KM}\} * \{\text{Per-KM Rate}\}$$

Detailed Pricing Grid with starting price and kilometer rate for each level

Tier	Type of transport	Base Fare (for the first 2 km)	Per-KM Rate (above 2 km)	Example: 7 km delivery fee
Level 1 (Express)	By foot / taxi / train	60 birr	15 Birr / km.	60 + (5 times 15) = {135 Birr}
Level 2 (Standard)	Bicycle / Motor scooter	80 birr	15 Birr / km.	80 + (5 times 15) = 155 Birr

Level 3 (Secure)	Selected engine/car	180 birr	25 Birr / km.	180 + (5times 25) = 305 birr
Level 4 (Cargo)	Suzuki / Isuzu (FSR)	600 birr	80 birr / km.	600 + (5times 80) = 1,000 birr

Master Tier Matrix

Tier Name (Tier)	Type of transport	Type of item	The highest price of the item	Time speed	Delivery price (origin)
Level 1 (Express)	By foot / taxi / train	Documents, clothing, mail and other services	Up to 3000 birr	2 - 4 hours	60 – 100 birr
Level 2 (Standard)	Bicycle / motorbike	Food, cake, medicine	Up to 7,000 birr	30 - 60 minutes	120 – 200 birr
Level 3 (Secure)	Selected engine/car	Phone, laptop, valuables	From 8000 to 100,000 birr	Direct	250 – 500 birr

Level (Cargo)	4	Suzuki / Isuzu (FSR)	Household /office goods, cargo	No (Insured)	At the appointed time	800 – 5,000+ Birr
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Section 5: Market & Competitor Analysis

The "Street Delivery" project is a part of an in-depth analysis of the market size, customer needs, and gaps in existing competitors' operations in Addis Ababa.

Although the delivery sector in our country is growing rapidly, high prices and lack of modernity in operations still leave a big gap. This section analyzes the current market situation as follows.

1. Market size and potential (Total Addressable Market - TAM)

Addis Ababa is a city with a population of over 6 million, with over 3 million daily transactions, and a rapidly growing e-commerce (online shopping) activity.

- **Total Market Capacity (TAM):**From daily activities in Addis Ababa**Over 100,000**Small transfers of goods, documents, food, and parcels.
- **Our Focus Market (SAM):**Between small online businesses (Social Media Vendors), offices, pharmacies, and individuals, there are over**Over 50,000**Small delivery activities.
- **Primary Market Objective (SOM):**On the 1st day of the year**1,200 orders (2.4%)**In the 5th year of holding**25,000 orders (50%)**Raising.

2. Customer Segmentation and Demand

Customer type	Profile and character	Their main concern (Pain Point)	On-street delivery solution

Online Merchants (Instagram/Telegram Sellers)	Small merchants selling clothing, shoes, cosmetics, and electronics	The high cost of delivery is a barrier for their customers (200-300+ birr)	Big discount on delivery Increase their sales by doing (in steps 1 and 2)
Regular offices and institutions	Documents, stamps, receipts, and letters sent daily	Hiring your own messenger is expensive and unreliable.	Express and reliable Service that works with OTP authentication
Individuals and students	Those who want to send forgotten items, food, medicine, or research documents	Paying the full price of a motor delivery to send a small item	By travelers on foot/by taxi Very low rate payable by cash/telebank

3. Competitor Landscape Analysis

Currently, competitors in the city can be divided into three main categories:

1. Traditional motorcycle deliveries] → (Example: Deliver Addis, BeU, Tikus)
2. Cargo and postal companies] → (e.g. Ethiopian Post, DHL)
3. Ride-hailing services] → (e.g. Ride, Feres, ZayRide)

Competitive Comparison Table

Competition criteria	Traditional motor deliveries	Ride-hailing taxis (Delivery)	Street Delivery (us)

Average delivery price	Heavy/expensive (250 - 400+ birr)	Very expensive (300 - 600+ birr)	Very cheap (150 - 250 birr)
Assets	They buy/hire motorists/vehicles.	The driver's car is on fire.	No vehicle (Asset-Light)
Impact of traffic congestion	High (blocked on the road)	High (delayed by traffic jams)	Low (pedestrian/cyclist/train use)
Environmental pollution (Carbon Footprint)	They emit high levels of carbon dioxide.	They emit high levels of carbon dioxide.	Zero/very low carbon emissions
Source of income for addressees	Full-time work for a few	Full-time drivers	Additional or full-time income for any traveler/student

4. Our Competitive Advantages / Unique Selling Propositions

1. **Cost Leadership:** Since the delivery people are already on foot, there is no additional fuel or time cost. This reduces the cost of delivery.
2. **Green Logistics:** By using existing people's journeys without additional motorbikes on the road, Addis Ababa's traffic congestion and air pollution will not increase.
3. **High Scalability:** It is possible to build a network effect of thousands of contacts in a few months.
4. **Technology adapted to local practices:** It is perfectly suited for the Ethiopian market as it is based on Faida Digital ID, Telebr/Chapa payments, and Landmark address processing.

Part 6: Route-Matching Algorithm

This is the heart of the "street delivery" project. This algorithm should not be based on a simple GPS connection, but should be based on the unique road layout, transportation culture, and network conditions of Addis Ababa.

This system, which combines the needs of the shipper and the direction of the traveler (addressee) and connects the right person with the right order, is explained in relation to the actual situation of our country as follows:

1. Localized Routing based on landmarks

The use of standard street names and house numbers in Addis Ababa is poor. Google Maps alone is not enough.

- **The actual situation of our country:**The address is called Summit, Jemo, CMC, Piazza, or "Down the taxi rank" in famous places (Landmarks).
- **The algorithm solution:**The system divides the city into small zones through "geo-fencing." The algorithm connects orders not only by map pins, but also by calculating the route from the addressee's current location to the neighborhood they are traveling to.

2. Route & Mode Matching

The algorithm takes into account the route the addressee takes and the type of transportation they use.

- **The actual situation of our country:**Most people travel by taxi (higgat/minibus), train, or on foot. These modes of transport cannot carry large items.
- **The algorithm solution:**When the system receives a request, it compares two things:

1. **Route Alignment:** If the address is going from 4 km to the intersection, the algorithm will only give it objects that are on the line (for example, Urael or Kazanchis).
2. **Item size:** For people traveling on foot or by taxi (level 1), it only allocates light documents and envelopes; for cargo (level 4), it only selects trucks.

3. Trust & Value Scoring

Anyone is not allowed to deliver whatever goods they want.

- **The actual situation of our country:** The risk of online fraud and theft is high.
- **The algorithm solution:** The system assigns each addressee a "Trust Rating." The higher the value of the item (e.g., a laptop), the algorithm assigns the item only to Level 3 addressees who have a Fayda ID, have been in the system for a long time, and can provide escrow.

4. Offline & Traffic Resilience

- **The actual situation of our country:** The network may be down or data speeds may be reduced. There is also heavy traffic congestion that fluctuates by the hour in areas such as Medan, Bole or Piazza.
- **The algorithm solution:** Even if mobile data is interrupted, the system will receive the address's latest location update via SMS (SMS/USSD). The algorithm also calculates traffic congestion for car users (using Google Traffic API), prioritizing routes during peak hours. **For those who walk or ride a bike** Gives.

The Algorithm Scoring Model

The algorithm uses the following match score to select the correct address for a command:

$$\text{MatchScore} = W_1 / \{\text{Distance}\} + (W_2 * \text{RouteAlignment}) + (W_3 * \text{TrustLevel}) - (W_4 * \text{TrafficPenalty})$$

Explanation:

- **W_1, W_2, W_3, W_4:** Each criterion has a weight (Weights).
- **Distance:** The distance from the address to the object (the smaller the distance, the higher the score).
- **Route Alignment:** How similar are the address access and the object access?
- **TrustLevel:** Address star rating and legitimacy verification.
- **Traffic Penalty:** Traffic congestion fine if by car (if by foot/bike this will be zero).

The recipient with the highest Match Score will be sent an Accept message on their phone application.

Section 7: Security & Chain of Custody

This is a critical point of failure or key success for implementing Crowdsourced Logistics in Addis Ababa.

The main threat to our country's market "**What if the addressee disappears with the item?**" "**What if the item breaks or needs to be replaced?**" And "**What if I don't know who the addressee is?**" These are the questions. Therefore, this section analyzes five (5) strict security safeguards that 100% address these threats, adapting them to the realities of our country.

1. Photo/Video Verification

Delivery disputes in our country arise due to controversial issues such as "the item is already broken/damaged" or "the recipient did not receive it."

- **Chain of Custody:**

1. **When receiving goods (Pick-up):**When the recipient receives the item from the sender, he or she takes a photo or a short 360-degree video of the item using a phone application.
 2. **When delivering goods (Drop-off):**When the item arrives at its destination, the delivery person takes a photo of the item and the recipient again through the app.
- **The benefit:**An irrefutable digital proof of delivery is created that shows the condition of the item before it was shipped.

2. 2-Factor OTP Code System

It is difficult to implement a paper signature in our country to ensure that the item reaches the right person (since signing on the phone is not convenient).

- **The procedure:**
 - The system generates a 4-digit secret code (OTP) that is sent only to the recipient's phone from the moment the item is shipped.
 - The recipient only enters the recipient code on the application when they arrive at the location."**The goods have arrived**"The confirmation is valid and the service fee is released to the addressee.
- **The benefit:**The addressee cannot lie by saying that he did not deliver the item, nor can the sender deny that the item has not arrived.

3. National Digital ID (Fayda ID) and Identity Verification

Since it is difficult to trust an unknown person in Addis Ababa, address registration must be very strict.

- **Our country's situation and the solution:**
 - Each addressee must be registered in the system before being registered."**Fayda" Digital ID (Fayda ID API)**Fingerprint and facial recognition (Biometrics) are verified.

- Additionally, the system will capture the Kebele ID, immediate family address, and card ID image.
- The application requires the recipient to show their face (Live Selfie Verification) every day before starting work.

4. Digital Security and Wallet Escrow/Hold System

This system is the main method of preventing theft of high-value items (such as phones, laptops, clothing).

- **The procedure:**

- When a recipient accepts a phone worth 10,000 birr for delivery, the money in their mobile banking or Telebirr Wallet account is transferred to the system.**Holds (holds)::**
- Once the item arrives safely at the recipient's doorstep and is verified with an OTP, the blocked funds will be released immediately.
- If the recipient loses the item with the item, the blocked money will be paid directly to the owner of the item as compensation.

5. Insurance Policy for High-Value Goods

It is a procedure that provides additional peace of mind for cargo and high-value items (Level 3 and Level 4).

- **Our country's situation and the solution:**

- In collaboration with leading local insurance companies, the application "**Micro-Insurance**" It is included.
- While the sender pays a very small fee (e.g. 0.5% - 1% of the value of the item) for the item he sends, if the item is broken, damaged, or suffers a catastrophic accident on the way, he is fully compensated by the insurance company.

Summary Chain of Custody Flow

1. Fayda ID verified]
- ↓
2. The price of the item is held in Wallet (Escrow)]
- ↓
3. Photo/video of the item taken (Pick-up)
- ↓
4. GPS Live Tracking
- ↓
5. Confirmed by the recipient's OTP code (Drop-off)]
- ↓
6. Payment has been released to the addressee]

This practice not only demonstrates human loyalty, but also **Technological and financial infrastructure** Since it makes it safe to use, it can be fully implemented and successful in the actual conditions of our city.

1. **Registration filter:** National Digital ID (Fayda ID), District ID, Police Character Verification and 1 Government/Legal Business Guarantor.
2. **3-step photo verification:**
 - *Photo 1:* When the sender loads.
 - *Photo 2:* When the addressee receives the item (to determine whether the item was already broken).
 - *Photo 3:* When the recipient receives it.
3. **Two-step OTP:** If the recipient does not provide the 4-digit code received on the phone to the addressee and does not complete the application, the contract will not be completed and the money will not be released to the addressee.

Section 8: Legal Structure and Taxation (Legal Compliance)

Implementing digital platforms and crowdsourced logistics in Ethiopia requires adhering to strict legal and tax frameworks. This section analyzes how the “street delivery” project can comply with our country’s commercial, legal, communication, and tax laws and maintain its legality.

1. Business Law and Corporate Structure

- **Legal form of the organization:**The organization is registered under the Ethiopian Commercial Code."**Private Limited Liability Company" (PLC)**It is recorded as .
- **Business License Categories:**The following licenses are issued by the Addis Ababa Trade Bureau and the Ministry of Innovation and Technology (MInT):
 1. **Software development and digital platform services** (*Software Development & Digital Platform Services*)
 2. **Freight and courier agency/forwarding** (*Freight Forwarding & Courier Brokerage Services*)
- **Fayda ID Linking Policy:**The API Data Access Agreement with the National Information Security and Fayda ID Agency will comply with the Data Privacy Act.

2. Courier Legal Status

The pedestrians, students, and drivers participating in this project are not regular employees of the organization.

- **Independent Contractor Agreement:**
 - Each address is digitally verified before being registered on the application."**Freelancer Agreement"**It will be signed.
 - The addressees have complete freedom to choose their own time, route, and transportation option.

- The company does not set working hours, minimum travel limits, or monthly salaries; they are paid only based on the number of deliveries they make.
- **Limitation of Liability:** If travelers are found carrying illegal items (e.g., drugs, weapons, illegal currency) or if they encounter an accident during their trip, the organization will prepare a legal framework that outlines their personal liability according to the contract.

3. Tax Compliance

According to the law of the Ethiopian Ministry of Revenue and the Addis Ababa Revenue Bureau, the tax process is carried out in three directions:

3.1. Value Added Tax (VAT - 15%)

- The organization will be registered as a 15% VAT (Value Added Tax) collector.
- The 15% VAT collected on each delivery payment is remitted to the Ministry of Revenue through a monthly VAT return.

3.2. Withholding Tax (2%)

- **Our country's law:** When an organization makes a payment to an independent contractor or service provider **2% Withholding Tax** He is obliged to reduce it and submit it to the government.
- **Automatic procedure:** When the application releases the delivery fee (82% of the amount) to the wallet for the addressee, it automatically deducts 2% withholding tax and deposits it in the organization's Hold Account. This money is remitted to the Addis Ababa Revenue Bureau every month.

3.3. Corporate Profit Tax (Corporate Profit Tax - 30%)

- The company's annual net profit **30% business profit tax** It will be paid to the Ethiopian government.

4. Forbidden Goods & Security Compliance

The following procedures shall be established by law to ensure the security situation in Addis Ababa and to enforce the laws of the country:

1. **List of Prohibited Items:**

- Weapons, explosives and ammunition.
- Illegal drugs, sedatives, and over-the-counter medications.
- Unauthorized cash, gold, and precious metals.

2. **Right to inspect and receive:** The recipient has the legal right to open and inspect the shipment before receiving it. If the shipment is suspicious, he/she has the right to immediately reject it and report it on the application.

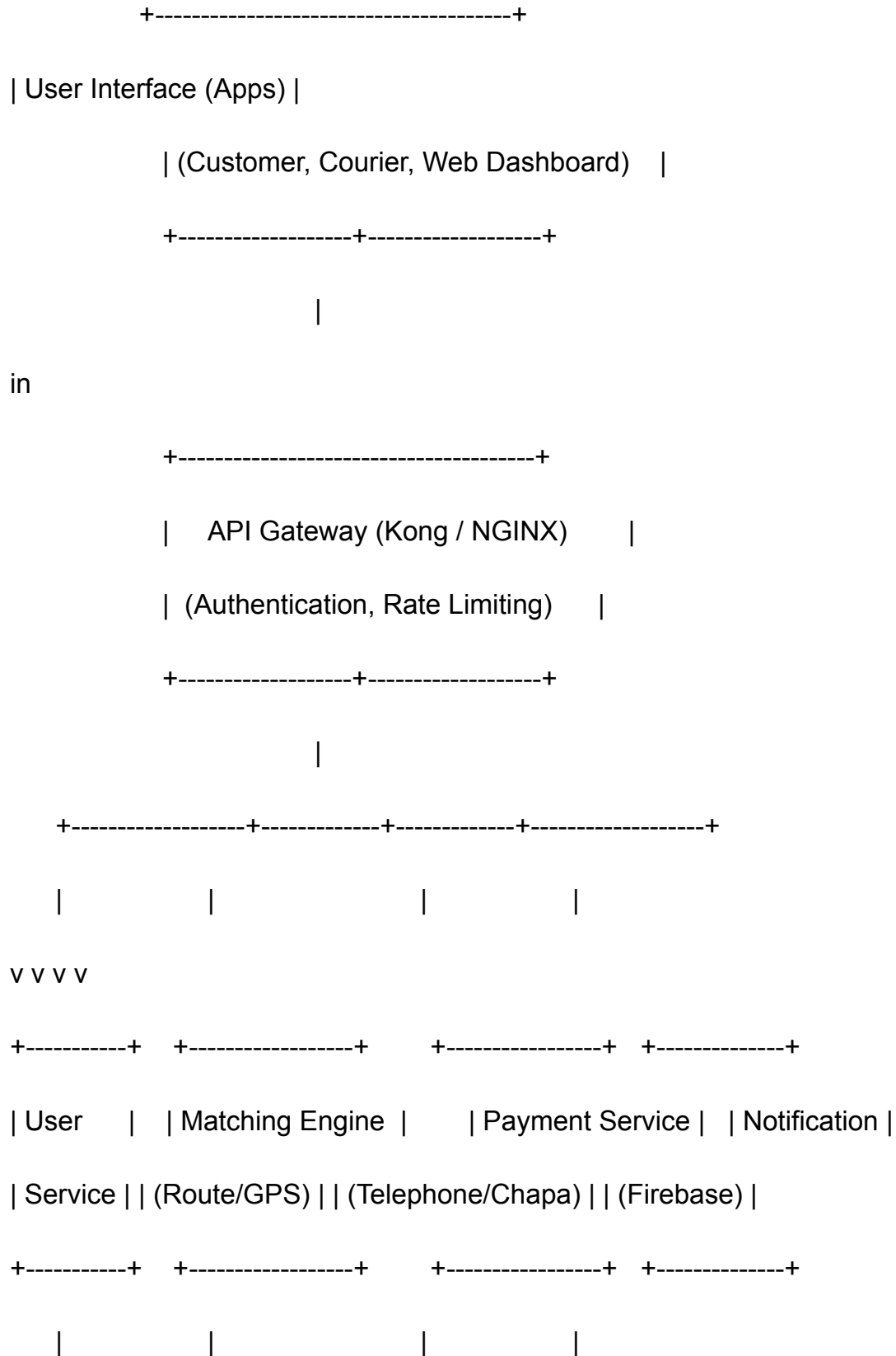
3. **Cooperation with police and security agencies:** Since the application contains the addressee's live GPS location and Fayda ID information, it is designed to provide the information to security agencies in accordance with legal procedures if illegal activity is suspected.

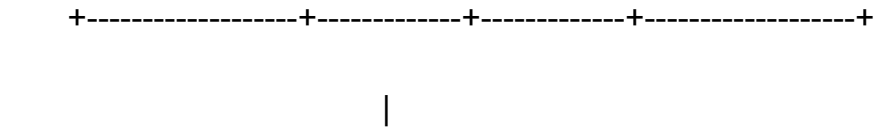
Section 9: Technical Software Architecture (Technical Specifications)

- **Frontend:** Flutter (iOS, Android እና Web)::
- **Backend:** Node.js (NestJS) microservices structure.
- **Database:** PostgreSQL + PostGIS (for spatial and map calculations) + Redis (for fast cache).
- **Payments:** Telebirr API, Chapa, CBO, CBE Birr payment gateways.

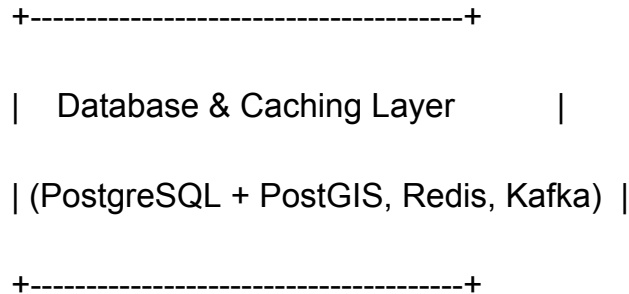
9.1. System Architecture

The platform is built by **Event-Driven Microservices Architecture**. This means that each part of the system (e.g., payment section, card matching section, user registry) operates independently, so even if one part shuts down, the entire application does not stop.





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9.2. Technology Stack

The app works with three simple clicks:

- **To the sender:**Select the type of item → Enter the destination address → Check the final price and order.
- **To the addressee:**Enter their line on Where are you going?; the system will only show items on that route.
- **For those who cannot use the application:**You can order through the free Telegram Bot and Call Center.

1. Frontend / Client Applications

- **Cross-Platform Mobile App (Flutter):**Applications for senders (Customer) and addressees (Courier) are available at**Flutter (Dart)**This allows for fast development on both Android and iOS phones with a single codebase.
- **Web Dashboard (React.js + TypeScript):**A fast and modern web page for the company's administrators (Admin Panel) and call center staff.

- **Telegram Bot (Node.js):**A bot that allows customers who don't have a smartphone or don't want to download an application to order goods just through a Telegram message.

2. Backend and Engineering (Backend Services)

- **Core API Framework (Node.js / NestJS):**To handle high volumes of map and command calls (Asynchronous I/O Operations) within seconds.
- **Real-time Communication (Socket.io / WebSockets):**To show the addressee's GPS movement and the status of the item directly on the map (Real-time Tracking) to the sender and recipient.
- **Message Broker (Apache Kafka / RabbitMQ):**A data queue controller that sequentially processes thousands of OTP SMSes, payment confirmations, and order calls.

3. Database & Caching

- **Primary Database (PostgreSQL + PostGIS Extension):**The main users, Order and tariff database. **PostGIS** The extension helps calculate geospatial data and road distances at high speed.
- **Caching Layer (Redis):**It relieves the pressure on the main database by temporarily storing the current location (Live GPS Locations) and starting values of active addresses.
- **Object Storage (MinIO / AWS S3):**Secure cloud storage for storing copies of Fayda ID, photos of recipients, and photographs of legal documents.

9.3. Smart Connectivity and Mapping (Location & Routing Engine)

1. **Mapping Provider:**Mainly **OpenStreetMap (OSM) / OSRM (Open Source Routing Machine)** Reduce GPS costs by 90% by deploying on your own server; and optionally use Google Maps API connectivity to get a better view of the city.

2. **Distance and location calculation (Geofencing):**Addis Ababa into the system
120 Logistics HexagonsDivide and organize.
3. **Route-Matching Rules:**
 - When the addressee registers the origin and destination of the trip, the system searches for items within a 300-meter radius.
 - The angle of the address with respect to the destination of the object should not exceed 20 degrees (20°).

9.4. Payment Integration & Escrow

- **Digital Payment Gateways: Telebirr API, Chapa, CBO (Coopbank) and CBE Birr**They connect directly to the app.
- **In-App Wallet & Escrow:**
 - When the customer makes a payment, the money does not go directly to the recipient; instead, it is transferred through the system.**Escrow Wallet**It is blocked inside.
 - The money is released into the recipient's wallet only when the recipient provides the 4-digit OTP secret number to the recipient and the recipient fills it in the app.
- **Automatic commission and tax:**The system of the organization**18% commission**And legal**2% Withholding Tax**It instantly deducts and transfers the remaining 80% to the recipient.

9.5. Security, Information Protection and Resilience (Security & Resilience)

1. **Data Security (Data Encryption):**All information exchanges are**TLS 1.3/HTTPS**Protected, security passwords and banking information are**AES-256**Algorithms are stored encrypted.
2. **Authentication:**It is equipped with two-factor authentication (JWT Tokens + Phone SMS OTP).
3. **Internet Outage Recovery (Offline/Low-Bandwidth Mode):**

- In areas where the internet is down, addressees can send delivery photos and OTP codes via SMS (**SMS Gateway Integration**) You can send.
- The application stores the data it sends in the phone's local SQLite storage and automatically synchronizes it with the main server when the internet is restored.

9.6. Software Development and Deployment Pipeline (CI/CD Pipeline)

[Code Commit (GitHub)] → [Automated Testing] → [Docker Containerization] → [Kubernetes Deployment]

- **Cloud Infrastructure:** The application is **Docker Containers** Wrapped in **Kubernetes Cluster** It is installed on a Cloud Infrastructure, which allows the server to automatically expand its capacity (auto-scaling) if the number of orders suddenly increases.

9.7. The system is structured around three (3) main components and a central admin dashboard:

- **Mobile Apps (iOS & Android):** Flutter (to set up the same code structure for both)
- **Back-End API:** Node.js (Express.js) or Python (FastAPI)
- **Database:** PostgreSQL h **PostGIS Extension** (For quickly calculating map and geographic data/Location Data)
- **Real-time Engine:** Socket.io or Firebase Realtime Database (for live GPS tracking)
- **Cloud Infrastructure:** AWS or DigitalOcean (Docker Containers)

9.8 Functional Requirements for the Three Applications

A. Customer / Sender App

1. **Registration and Account:** Register with phone number (OTP).
2. **Order Creation:**
 - Enter the item type, weight, and estimated value.
 - Selecting a pick-up and drop-off location via a map or a famous landmark.
 - Choosing one of the four service levels (Express, Standard, Secure, Cargo).
3. **Price Calculation Engine:**
$$\{\text{Total}\} = \{\text{Base Fare}\} + \{\text{Distance in KM}\} * \{\text{Rate per KM}\}$$
4. **Live GPS Tracking:** See the delivery person moving with the item live on a map.
5. **OTP Verification:** Seeing a 4-digit secret code given to the addressee when the item arrives.

B. Courier / Traveler App

1. **Identity Verification:**
 - Enter your National Digital ID (Fayda ID) number.
 - Face recognition (Live Selfie / Liveness Detection).
2. **Route Intent Input:**
 - Enter the address's current location and destination (Example: Kebole → Memande).
3. **Route-Matching System:**
 - Show only orders on the address's route in Notification.
4. **Proof of Pick-up & Delivery:**
 - Making it mandatory to take a photo on the application when receiving and delivering the item.
5. **Digital Wallet (In-App Wallet):** View the money made and transfer it to your bank/telebank immediately (Payout).

C. Central Admin Dashboard (Admin Web Portal)

1. **User & Courier Management**:-Verify and Approve/Block the addressee's ID, Fida ID status.
2. **Order Monitoring**:-Live tracking of all active orders in the city on a map.
3. **Dispute Resolution System**:-If an item is lost/damaged, a decision can be made based on photos of the sender and recipient, as well as GPS history.
4. **Financial Ledger & Reports**:-18% commission collection, 2% Withholding Tax reporting.

9.9 Third-Party API Integrations

External APIs that the professional must include:

1. **Fayda Digital ID API**:-To directly verify the identity of the addressees.
2. **Telebirr & Chapa Payment APIs**:-
 - Checkout
 - To implement automatic wallet escrow/hold.
3. **Google Maps API / OpenStreetMap (Nominatively)**:Distance Matrix and address search.
4. **SMS Gateway (e.g., AfroMessage / Telebirr SMS)**:-To send OTP codes.

9.10 Security and Data Requirements

1. **Role-Based Access Control (RBAC)**:-Sender, addressee, and admin data should be separated.
2. **Data Encryption**:-Passwords and sensitive information should be hashed (bcrypt); API connections should be over HTTPS/TLS.
3. **Escrow Logic**:-Payments should not be released to the recipient until the item has been received and verified with an OTP.

Section 10: Business Model & Revenue Streams

1. **Platform Commission**:
 - Items from Level 1 and Level 2:**20% commission**

- From Level 3 (Secure) items:**15% commission**
 - Items from Level 4 (Cargo):**12% commission**
2. **Packaging Revenue for the future:**Sales of cardboard boxes, specialty envelopes, and bubble wrap.
 3. **Corporate Subscriptions:**Post-paid payment system for banks, law firms, and merchants.

The "street delivery" business model is driving the growth of digital payments in Ethiopia.**Telebirr** (CBE BirrAndPlate), which takes into account online business activity and the city's existing transportation conditions."**Asset-Light Commission & Service-Based Model**" is it.

The company has very low start-up and operating costs because it does not purchase vehicles, engines, or warehouses. Revenue is generated through commissions from each delivery transaction and additional services.

1. Primary Revenue Streams

1.1. Platform Commission Fee (Transaction Commission - 18%)

- **Operating conditions:**The total payment for each completed delivery service is based on**18% commission**Remains; the rest**82%**The funds will be deposited directly into the recipient's (traveler's) wallet.
- **Example (from Addis Ababa):**
 - If a student delivers documents or clothes from Bole Medhanealem to Medanka (4.5 km).
 - Total fare = 60 birr (starting point) + (4.5 km × 15 birr) =**127 birr**
 - **Platform revenue (18%) = 22.95 Birr**
 - **The share of the address (82%) = 104.55 birr**(The student/traveler will be responsible for the taxi fare).

1.2. Security & Risk Mitigation Fee

- **Operating conditions:**For Level 3 (valuable items: phones, laptops) and Level 4 (cargo) items, the system will require 2-step OTP, Fayda ID verification, and additional insurance/compensation funds when shipping.**2% - 3% security service charge**It will be added to the fee.

1.3. B2B Corporate and Online Merchant Membership (Merchant Subscription Plans)

There are thousands of e-commerce merchants on Telegram, Instagram, and TikTok. There are monthly plans for these merchants:

- **Basic Plan (Free):**Standard 18% commission payout.
- **Business Plan (Business Plan - 2,000 Birr/month):**The commission drops to 14% + Priority Matching + Telegram Bot Automation.
- **Premium Plan (Enterprise Plan - 4,500 Birr/month):**Commission drops to 10% + special API connection for large pharmacies, supermarkets and law firms.

1.4. In-App Ads & Featured Merchants

- **Operating conditions:**Online shops and restaurants will be featured as "Featured Merchants" on the app's front page and will be advertised to users, earning up to \$100 per month.**500 to 2,000 birr**Collecting payment.

2. Dynamic Pricing Structure

The pricing for the 4-Tier service in Addis Ababa city is as follows:

Service Price}= {Base Fare} + ({Distance in KM} * {Per-KM Rate}) + {Surge Factor}

Service level	Base Fare	Per kilometer (Per KM)	Target item type	Recommended mode of transportation

Level 1: Express	60 birr	15 Birr/km	Documents, mail, clothing	By foot, taxi, public train
Level 2: Standard	80 birr	15 Birr/km	Food, medicine, gift	By bicycle, electric scooter
Level 3: Secure (Secure - High Value)	180 birr	25 Birr/km	Phone, laptop, valuables	Strictly by courier + Escrow
Step 4: Cargo	600 birr	80 birr/km	Home/Office Furniture	Suzuki (Carry) / Isuzu

- **Surge Pricing:** During heavy rains (winter) and holiday seasons (New Year, Easter, Easter), the fee will automatically increase by 15% - 25%, which encourages delivery drivers to work in the rain and crowds.

3. Digital Payment Connectivity and Money Transfer (Financial Infrastructure)

A system designed to be in line with the cash and digital payment culture in our country:

- **Digital Wallet Connection (Telebirr / Chapa / CBE Birr):**
 - Customers pay directly with Telebirr, CBE Birr, or their bank card.
 - The payment is made through the system. **Escrow (temporary escrow)** It stays inside.
 - The item arrived and the recipient received the message on his phone. **When verifying OTP** 82% will be credited to the addressee's wallet immediately; 18% will be credited to the company.
- **Cash on Delivery (COD) Compensation:**
 - If the recipient wants to pay in cash, the addressee will receive the money.

- The company's 18% commission is automatically deducted from the addressee's in-app wallet. (For this, addressees must have a minimum deposit in their Wallet account).

4. Unit Economics Analysis (Unit Economics per Order)

An average **5 kilometers** If a Level 1 (Express) order is taken that covers:

Total gross merchandise value (GMV) per order = $60 + (5 \times 15) = 105.00$ Birr

Main summary: The organization will receive a 105 birr discount on every order. **Net Profit: 18.9 Birr** He gets it. In the 1st year, when 1,200 commandments are implemented per day, **22,680 birr (680,400 birr per month)** It generates net profit.

Section 11: Go-to-Market Strategy (GTM)

The "Street Delivery" marketing strategy is designed to quickly build a large customer and address base (Network Effect) at low cost. It takes into account the current digital media usage in Addis Ababa, the growth of online commerce (Social Media Commerce), and traditional neighborhood/word of mouth advertising (Word of Mouth) conditions. **4-step strategy** It is analyzed as follows.

1. Two-Sided Network Acquisition

On the Crowdsourced Logistics platform **Senders (Demand) And Addresses (Supply)** They are equally important.

- **Supplier Notification Method:**

- **University students:** Register at Addis Ababa University (AAU), Kotebe, and private colleges to "earn money while you go." Students turn their daily commute from campus to home/work into income.
- **Taxi/Bicycle/Train Passengers:** Registering addresses in brochures and with campaign staff at major transportation hubs (e.g., Medaka, Bole, Arada Giorgis, Stadium, Saris).

- **Sign-Up Bonus:**The first 1,000 subscribers will receive a credit on their Wallet account.**100 Birr starting bonus** give away.
- **Vendors/Senders' Notification Method:**
 - **Telegram/TikTok Merchants Connection:**More than 50% of unorganized online commerce in Addis Ababa is conducted on Telegram channels and TikTok. For these merchants"**Dramatically reduce your delivery costs**"Making a B2B deal.
 - **Free Trial Opportunity (Free First Delivery):**Offering merchants a free delivery promotion for the first 3 items.

2. Digital Marketing & Influencer Campaign

- **TikTok and Instagram content (TikTok/Instagram Short Videos):**
 - **Educational and entertaining content:**Making real and short videos that say, "How did a student manage to cover the cost of the taxi and lunch by taking a taxi from 4 kilometers to the intersection, delivering 150 birr worth of goods?"
 - **Influencer Referral:**A system where popular local content creators earn commission when customers sign up using a Referral Code.
- **Telegram Bot and Channel Strategy (Telegram Ecosystem Integration):**
 - Given the high usage of Telegram in Ethiopia, facilitating a system where customers can send items through Telegram Bot without installing the application.

3. On-Ground Activation

- **B2B Merchant Onboarding:**
 - By organizing small sales teams (Sales Reps),**Mercato (clothing and electronics), Bole Medhanealem, Medanka Marathon and CMC**Register with the application by visiting legitimate stores and shops in person in major shopping areas.
- **Zonal Phased Launch:**

- **Chapter 1 (Months 1-3):** Starting with just 3 high-traffic corridors:
 1. *Bole → Junction*
 2. *4 km → Piazza → Stadium*
 3. *Junction → 22 → Kasanchis*
- **Chapter 2 (Months 4-6):** Expand to the entire Addis Ababa (Jemo, Summit, Ayat, World Bank, Akaki).

4. Customer Loyalty and Referral Program (Retention & Referral Program)

- **"Refer-a-Friend" Program:**
 - When a customer or affiliate invites (refers) another new person, both of them receive **50 birr** You will receive delivery credit/bonus.
- **Loyalty Tiers for Sellers:**
 - Providing Discounted Flat Rates and Priority Matching for online merchants who ship more than 50 items per month.

Market Expansion Budget and Goal (GTM Key Metrics)

Measurement type	1st Year Goal (KPI)
Number of Addressees (Active Couriers)	5,000+ registered hikers
Number of Merchants/Exporters (Active Merchants)	1,200+ Telegram/Instagram and stores
Customer Acquisition Cost (CAC)	from 45 to 60 birr By customer
Organic Referral Rate	More than 35% of new customers come from referrals

Section 12: Comprehensive Financial Projection and Economic Analysis (Comprehensive Financial Projection: 2026–2030)

Type of expense	Estimated cost (in Ethiopian Birr)
Mobile Application (Android/iOS) + Admin Dashboard	450,000
Server, SMS Gateway, and API costs	100,000
Call Center and Telegram Bot	80,000
Legal registration, securities and office organization	70,000
Digital Marketing and Youth Registration	150,000
Total initial capital	850,000 birr

12.1. Market Capacity & Key Assumptions

1. **Addis Ababa Daily Delivery Demand (Addressable Market Volume):** Average daily in the city **From 50,000 to over 80,000** Transportation and delivery activities are carried out.
2. **Market share strategy (Market Penetration Target):**
 - **Year 1:** Average per day **1,200 Commandments** (Having a ~2% share of the city's delivery market).
 - **Year 2:** Average per day **3,500 Commandments** ::
 - **Year 3:** Average per day **8,000 commandments** ::
 - **Year 4:** Average per day **15,000 commands** ::
 - **Year 5:** Average per day **25,000 orders** (Addis Ababa + Adama/Hawassa/Bahir Dar).
3. **Blended Average Order Value (AOV):**
 - Based on the average calculation across all levels (levels 1 to 4) **Average payment per order = 160 birr** ::
4. **Average Platform Commission: 18%** ::
5. **Inflation Adjustment:** On average every year **15%** Tariff and operating cost adjustments will be made.

12.2. Initial CAPEX Breakdown

The initial capital required to build the initial technology infrastructure and start working **850,000 Ethiopian Birr** is it.

+-----+

| Initial Capital Expenditure (CAPEX Breakdown) |

+-----+

| 1. Mobile Application and Software Development (iOS, Android, Web, Driver App) |
450,000 Birr |

| 2. Server, Cloud Infrastructure (AWS) and SMS Gateway Pre-payment | 100,000 Birr |

| 3. Call Center Equipment and Telegram Bot Setup | 80,000 Birr |

| 4. Legal Registrations, Contract Documents, Trademarks and Licenses | 70,000 Birr |

| 5. Opening Digital Marketing, Traveler/Youth Registration and Promotion | 150,000 Birr |

+-----+

| Total Initial Investment Requirement | 850,000 Birr |

+-----+

12.3. 5-Year P&L Statement

(All numbers are in Ethiopian Birr)

Item list	Year 1 (Year 1)	Year 2 (Year 2)	Year 3 (Year 3)	Year 4 (Year 4)	Year 5 (Year 5)
Daily Orders	1,200	3,500	8,000	15,000	25,000
Annual Orders	438,000	1,277,500	2,920,000	5,475,000	9,125,000

Gross Market Value (GMV)	70,080,000	235,060,000	617,872,000	1,333,710,000	2,557,108,000
Platform commission income (18% Commission)	12,614,400	42,310,800	111,216,960	240,067,800	460,279,440
Additional Revenues (Packaging, Ads & Corporate)	950,000	3,500,000	9,500,000	22,000,000	42,000,000
Gross Revenue	13,564,400	45,810,800	120,716,960	262,067,800	502,279,440
Direct Operating Expenses (COGS / Direct OPEX)					
- Cloud Server, SMS Gateway & Maps API	950,000	2,200,000	5,200,000	9,800,000	16,500,000

- Payment gateway fees (Telebirr/Chapa 1-1.5%)	840,960	2,820,720	7,414,464	16,004,520	30,685,296
- Compensation and Damage Cover Fund (Risk Fund 2%)	252,288	846,216	2,224,339	4,801,356	9,205,589
Total direct costs (Total COGS)	2,043,248	5,866,936	14,838,803	30,605,876	56,390,885
Gross Profit	11,521,152	39,943,864	105,878,157	231,461,924	445,888,555
Administrative and Marketing Expenses (SG&A)					
- Employee salaries (IT, Customer Care, Operations)	2,800,000	6,500,000	14,000,000	25,000,000	40,000,000

- Digital and Field Marketing	1,800,000	4,500,000	11,500,000	22,000,000	35,000,000
- Office rent, utilities and administrative costs	600,000	1,400,000	2,800,000	5,000,000	8,000,000
Total SG&A	5,200,000	12,400,000	28,300,000	52,000,000	83,000,000
Earnings before tax (EBITDA)	6,321,152	27,543,864	77,578,157	179,461,924	362,888,555
- Business Profit Tax (30% Profit Tax)	1,896,346	8,263,159	23,273,447	53,838,577	108,866,567
Net Profit	4,424,806	19,280,705	54,304,710	125,623,347	254,021,988

12.4. Key Financial Metrics & ROI

1. Payback Period

- **Formula:** {Payback Period} = {Initial CAPEX}/{Year 1 Net Profit}
- **Calculation:**{850,000 { Birr}}/{4,424,806 { Birr}} = 0.19 { Years}

- **Explanation:**The project started based on the new market demand calculation.**2 months and 10 days**He will fully recover the initial capital of 850,000 birr he invested in it.

2. Internal Rate of Return (IRR)

- **5-Year Projected IRR: 185%**
- **Explanation:**The difference between the amount of investment and the amount of profit the organization earns is very high due to the fact that the organization uses the capacity of others without purchasing its own vehicles (Asset-Light Platform).

3. Socio-economic payment (Gig Workers Income)

- **Paid to addressees once a year: 57.4 million birr**
- **5 per year payable to addressees: 2.09 billion birr**
- **Social benefit:**More than 2 billion birr will be distributed directly to students, pedestrians, and youth moving around the city every day in the form of payments.

Section 13: Risk Analysis and Mitigation Framework

1. Security & Fraud Risk

Current situation and threats (Addis Ababa Context)

In our city, there has been a recent surge in the movement of electronic items (phones, laptops), office documents, and gold/jewelry. If a traveler loses, misplaces, or reports their belongings as stolen, the traditional practice of going to police stations to search for people can damage the business's reputation and lead to loss of customers.

Mitigation Strategies

- **Mandatory connection of Fayda ID:**Each addressee will be verified with a national biometric ID (Biometric Fayda ID). Their name, photo, and address will be directly linked to a government database, eliminating 99% of the need to hide items.
- **3-Step Live Photo Verification:**
 1. The addressee takes a photo when they receive the item (to ensure that the item was received undamaged).
 2. It constantly shows the status of the road via GPS.
 3. When delivering, he attaches a photo of the recipient and a map of the location where he received the item.
- **Wallet Escrow & Hold:**For general or expensive items (Level 3), the recipient must have a “hold” of the equivalent value of the item in their app wallet. The funds will be released when the item arrives safely.
- **2-step OTP secret number:**The payment will not be released to the addressee until the recipient provides the 4-digit number received by phone.

2. Traffic & Operational Delay Risk

Current situation and threats (Addis Ababa Context)

Addis Ababa experiences heavy traffic congestion during rush hour (especially between 1:30 and 3:30 in the morning and 11:00 and 1:30 in the afternoon). In addition, road maintenance, public taxi queuing, and holiday market activities are difficult to predict.

Mitigation Strategies

- **Multimodal Transport:**
 - Avoid delays in taxi queues and traffic
 - **Addis Ababa Light Rail Transit:**
 - **WalkingAndBicycles/electric scooters**They are encouraged to use it in a coordinated manner.
 - Short distances (1–3 km) will be covered more quickly by students and youth on foot or by bicycle than by car traffic.

- **Live Smart Routing (Dynamic Routing & Buffer Time):**
 - The system shows the shortest and shortest routes (shortcuts) to the address by looking at the current traffic conditions of the city.
 - By adding a 20-minute buffer time to the estimated arrival time, the customer is not disappointed.

3. Connectivity & Telecom Risks

Current situation and threats (Addis Ababa Context)

In our country, internet service is subject to disruption, network congestion, or server outages. If GPS service is interrupted, the addressee may lose track of where the item is being delivered.

Mitigation Strategies

- **SMS and Offline Mode:**
 - In places where the internet is unavailable or weak, the application communicates with the main server via SMS (SMS Gateway).
 - When the recipient enters the OTP password, the app stores the information on the phone and automatically synchronizes it with the system when the internet comes back (Data Sync).
- **Call Center Fallback:**
 - During times when the internet is completely down, customers and deliverers can call the toll-free call center (Short Code) and get their delivery address from the staff.
- **Dual Telecom API Integration with ET-Telecom and Safaricom:**
 - The application uses both Ethio Telecom and Safaricom networks equally, so if one is interrupted, the data exchange will continue on the other.

4. Social Acceptance & Cultural Risk

Current situation and threats (Addis Ababa Context)

In our culture, it is still unusual to give valuable items to a stranger, or to say, "I'll deliver something to someone I meet on the street." In addition, there is a risk that some recipients will view the recipient as an ordinary pensioner and belittle or behave inappropriately.

Mitigation Strategies

- **Rating & Review System:**

- Both the client and the sender give each other a rating (Rating 1-5 Stars) after each job.
- Any customer or host who exhibits unethical behavior will be immediately banned from the Platform.

- **Local Community Building (Community Hubs & Identification):**

- Attendants will be required to carry a character certificate, uniform (Vest/Reflector), and a recognized ID; this will create trust and respect among the public.

- **"On-Street" Brand Trust Campaign:**

- Promote mobility as a modern, green/eco-friendly, and youth-friendly work culture through social media and TV/radio.

5. Legal, Regulatory & Tax Risk

Current situation and threats (Addis Ababa Context)

The Addis Ababa Transport Bureau, the Ministry of Revenue, and postal agencies regularly update laws for digital applications and the economy. If the recipients are blocked for being "unregistered workers" or if there is a tax problem, they may be blocked.

Mitigation Strategies

- **Independent Contractor Agreement:**

- All addressees sign an electronic contract stating that they are "free service providers" under the law.

- **Automatic tax deduction (2% Withholding Tax System):**
 - The system deducts the legal 2% withholding tax from the income earned by the recipients and remits it directly to the Ethiopian Ministry of Revenue. This makes the organization free from tax risks.
- **Working in partnership with public transport:**
 - Meet legal requirements through preliminary discussions with the Addis Ababa Transport Bureau and the Postal Agency.

6. Financial & Inflation Risk

Current situation and threats (Addis Ababa Context)

Due to the inflation in the city, the price of taxis, fuel, and food is constantly increasing. If the fare is too low, drivers will lose interest in working; if it is too popular, customers will return to traditional taxis.

Mitigation Strategies

- **Dynamic Pricing Engine:**
 - The system automatically adjusts the payment based on the city's fuel prices, inflation, and surge pricing.
- **Disaster and Compensation Fund (2% Loss Recovery Fund):**
 - 2% of each payment is set aside for a damage and accident fund; this ensures that the customer is compensated within 24 hours of the loss or damage of an item.

Risk Matrix Summary

Type of threat	Chance of occurrence	Impact size	Key solution step
Theft/Theft of property	Low	High	Fayda ID + Wallet Hold + OTP

Traffic reporting	High	Medium	Use public transportation/walking
Internet outage	Medium	Medium	SMS (SMS Offline Mode) and Call Center
Loss of traditional faith	Medium	Medium	3-step photo, ID, and rating process
Tax and legal risks	Low	High	Automatic 2% Withholding Tax Discount in the System

Section 14: Implementation Roadmap

The Implementation Roadmap for the implementation of the “Street Delivery” project in Addis Ababa city should take into account the city’s specific conditions. This should take into account the rainy season (winter), holiday market activities (such as New Year, Easter, Easter), school opening and closing times, as well as government licensing and Fayda ID integration procedures.

[Chapter 1: Law and Technology] → [Chapter 2: Pilot] → [Chapter 3: Official Launch]
→ [Chapter 4: Expansion]
(Month 1 - 3) (Month 4 - 5) (Month 6) (Month 7 - 9)

Phase 1: Infrastructure, Legal Framework, and Software Development (Months 1 – 3)

This phase is when the legal foundation of the business is laid and the technology architecture is built.

1. Legal registration and government relations (month 1)

- **Business License and Tax Registration:** Issuance of a business license from the Addis Ababa Trade Bureau; Approval of the legal framework for the automatic system of TIN and 2% Withholding tax from the Ministry of Revenue.
- **Connecting to the national Fayda Digital ID:** Signing an API Integration Agreement with the National Information Security and Fayda ID Agency to directly verify the identity of addressees.
- **Payment gateway terms:** Finalizing Escrow Payments and Wallet integration agreements with Ethio Telecom (Telebirr), Chapa, and Commercial Bank of Ethiopia (CBE Birr).

2. Software and Map Development (2 – 3 months)

- **Building applications:** Completing the initial work of the Customer App, Courier App, Web Dashboard, and Telegram Bot.
- **Addis Ababa Map Localization:** Mapping of well-known neighborhoods (e.g. Bole, Medaka, 4 Kilo, Mercato, Piazza, Qaliti) and 120 sub-zonal divisions (Logistics Hexagons) on OpenStreetMap/Google Maps, taking into account the lifestyle of the city's population.
- **SMS and Offline Mode Testing:** Testing OTP and delivery procedures that work via SMS (SMS Gateway) during network outages.

Phase 2: Pilot Phase & Onboarding (Month 4 – 5)

This phase is a time to test the application in a small part of the city to ensure the robustness of the system and the readiness of the human resources.

1. Pilot Zone Selection (Pilot Strategy) (Month 4)

- **Location selection:** Where transportation demand and business activity are high
Bole-Medan-4km-four kilos Selecting lines for experimental control zones.

- **Targeted test customers:** Selling clothes, shoes, and electronics on Telegram and Facebook
- **50 Small Online Merchants (E-commerce Merchants)** Register for a free delivery service (Free Pilot).

2. Onboarding & Training (Month 4 – 5)

- **Registration for the first 300 subscribers:** Registering youth with Fayda ID in areas where there are a large number of job seekers and who commute daily by taxi/walk.
- **Local ethics and safety training:**
 - Customer reception and greeting culture (handing over goods with Ethiopian courtesy).
 - Use of security measures and photo/OTP system.
 - Training on the use of the city's fastest walking and taxi alternatives.
- **Addressees' clothing and identification:** Distributing reflective vests and digital IDs for safety and security.

Phase 3: Official Launch and Marketing Campaign (Month 6)

This phase is when the application is announced to the entire population of Addis Ababa and a major promotion is carried out.

1. Seasonal Marketing Strategy

- **Taking advantage of holiday market opportunities:** Taking advantage of the high demand for goods by launching before national holidays (e.g. New Year/Cross Day or Christmas/Easter Market).
- **Promoting a "nomadic" culture:** Broadcasting social promotions on TikTok, Telegram, Facebook, and FM radios that say "Earn money as you walk!" and "Let's reduce congestion in our city by walking!"

2. Promotional Pricing

- **First-Delivery Discount:** Allowing free commission service for new customers who download the app.

- **Courier Incentive:** Providing an additional 2-point commission-free delivery service for walkers who deliver more than 5 items per day.

Phase 4: Full Scaling (Months 7 – 9)

This is the phase where the application will expand its capabilities and expand to neighboring cities after it is fully operational in Addis Ababa.

1. Corporate Partnerships (B2B Partnerships) (Months 7 – 8)

- **Working with large organizations:** Signing long-term B2B contracts with pharmacies (for medicine delivery), law firms/offices that require document delivery, and large supermarkets.
- **Full launch of Level 4 (Cargo):** Extensively enrolling Suzuki and Isuzu drivers for moving household and office goods.

2. Geographic Expansion (Month 9)

- **Connections between Sheger and surrounding cities:** Integrate surrounding cities (Burayu, Sululta, Sebeta, Gelan, Bishoftu) that have close daily trade links with Addis Ababa into the system.
- **Expanding to regional capitals:** Starting pilot projects in Adama, Hawassa and Bahir Dar cities.

Master Roadmap Matrix

Phase	Key Activities	Target Timeline	Key Performance Indicators (KPIs)
Chapter 1: Law and Technology	Business License, Fayda ID API, Telebirr/Chapa Contract, Application Development	Month 1 – 3	The applications have been prepared and legal permits have been completed.

Chapter 2: Experimental implementation	Registration of 300 addresses, service of 50 online merchants, Bole-Medan test	Month 4 – 5	1,000+ test orders successfully completed
Chapter 3: Official Inauguration	Overall city launch, social media campaign, holiday market promotion	Month 6	Reaching 1,200+ orders per day
Chapter 4: Expansion	B2B corporate terms, cargo system launch, Sheger/surrounding cities connectivity	Months 7 – 9	5,000+ active subscribers and 3,500+ orders per day

Part 15. Conclusion

"Roadside Delivery" A low-cost, low-maintenance solution to address the high delivery costs and traffic congestion in Addis Ababa city. **850,000 birr**) demanding and rapidly profitable (**Payback period = 2 months**) is a new digital solution. The project is a socio-economically beneficial idea that not only supports the company's profitability, but also generates additional income for thousands of young people and travelers every day.

